

Mini Test 7 – Trigonometry

Name:

Date:

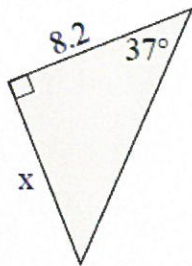
Total Marks: /44

Total Time: 45 minutes

Marks will be allocated for working out. Read the questions carefully.

1. Use the tan ratio to find the length of side x :

(2 marks)

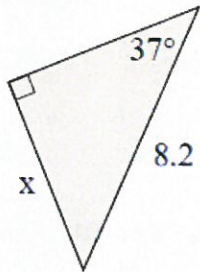


$$x = \tan 37^\circ \times 8.2 \quad \checkmark$$

$$= 6.18 \quad (2dp) \quad \checkmark$$

2. Use the sine ratio to find the length of side x :

(2 marks)

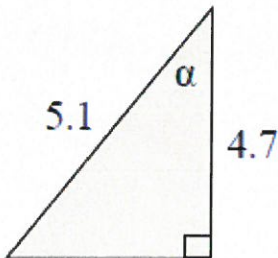


$$x = \sin 37^\circ \times 8.2 \quad \checkmark$$

$$= 4.93 \quad (2dp) \quad \checkmark$$

3. Use the cos ratio to find the size of angle α :

(2 marks)



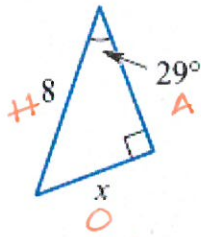
$$\alpha = \cos^{-1} \left(\frac{4.7}{5.1} \right) \quad \checkmark$$

$$= 22.84^\circ \quad (2dp) \quad \checkmark$$

4. For each of the following, find the value of x correct to two decimal places.
label sides (1 mark), choose the correct trigonometric ratio (1 mark),
show all working out before solving (2 mark)

(16 marks)

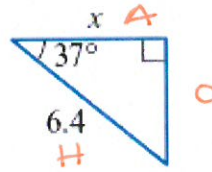
a



SOH ✓

$$\checkmark \sin 29 \times 8 = x \\ = 3.88 \text{ (2dp)} \checkmark \checkmark$$

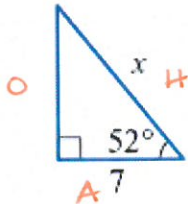
b



✓
CAH ✓

$$x = \cos 37 \times 6.4 \\ = 5.11 \text{ (2dp)} \checkmark \checkmark$$

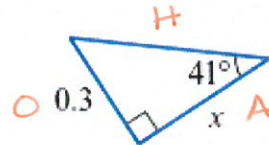
c



✓
CAH ✓

$$x = \frac{7}{\cos 52} \checkmark \checkmark \\ = 11.37 \text{ (2dp)} \checkmark$$

d



✓
✓ TOA

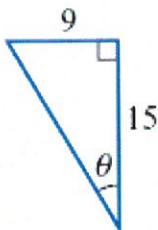
$$x = \frac{0.3}{\tan 41} \checkmark \checkmark \\ = 0.35 \text{ (2dp)} \checkmark$$

5. Find the value of θ to the nearest degree.

Choose the correct trigonometric ratio (1 mark), show all working out before solving (2 mark)

(6 marks)

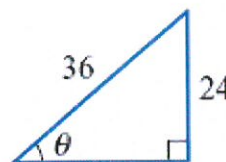
a



TOA ✓

$$\tan \theta = \frac{9}{15} \\ \theta = \tan^{-1} \left(\frac{9}{15} \right) \checkmark \\ = 30.96^\circ \\ = 31^\circ \checkmark$$

b



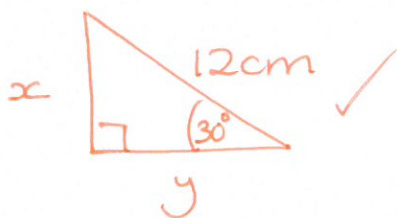
SOH ✓

$$\sin \theta = \frac{24}{36} \checkmark \\ \theta = \sin^{-1} \left(\frac{24}{36} \right) \checkmark \\ = 41.81^\circ \\ = 42^\circ \checkmark$$

* -1 total if not given to nearest degree.

6. Sketch a right angle triangle which has an angle of 30° and a hypotenuse of 12cm. Find the lengths of the other two sides. Round to the first decimal point.

(3 marks)



$$x = \sin 30 \times 12$$

$$= 6 \text{ cm} \quad \checkmark$$

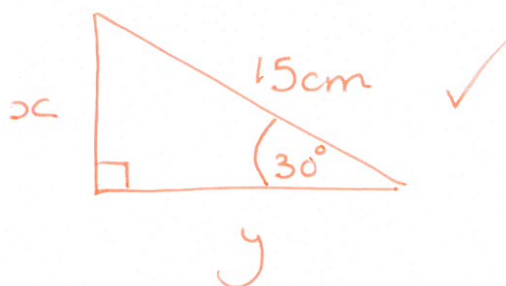
$$y = \cos 30 \times 12$$

$$= 10.4 \text{ cm} \quad \checkmark$$

7. Sketch a different right angled triangle with an angle of 30° where the hypotenuse is not equal to 12cm. Write on possible lengths of the sides.

(3 marks)

* other answers are possible.



$$x = \sin 30 \times 15$$

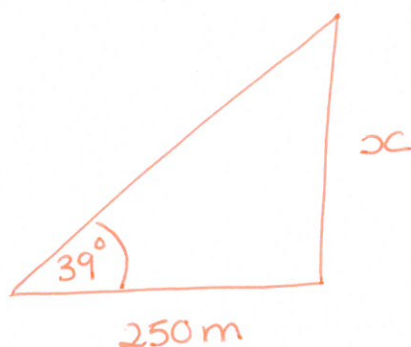
$$= \quad \checkmark$$

$$y = \cos 30 \times 15 \quad \checkmark$$

8. A surveyor measures the angle to the top of a peak from a point 250 m away to be 39° . How high is the peak?

(2 marks)

Hint: Draw a diagram.



$$\tan 39 = \frac{x}{250}$$

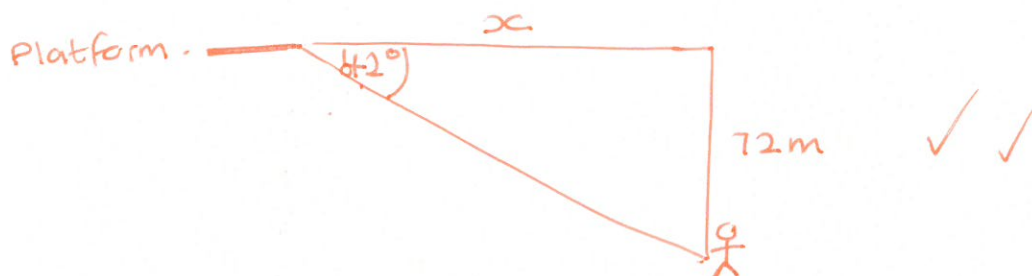
$$x = \tan 39^\circ \times 250$$

$$= 202.45 \text{ m} \quad \checkmark$$

(2dp) $\checkmark$

9. At **72 metres** in height, Gloucester Tree in Pemberton, is the world's second tallest fire-lookout tree (second only to the nearby Dave Evans Bicentennial Tree), and visitors can climb up to a platform in its upper branches for a spectacular view of the surrounding karri forest. The angle of depression from the top of the platform is 42° when looking to where someone stands on the ground. (4 marks)

a. Draw an appropriate diagram that shows all of the relevant information from the scenario above.



b. Calculate the distance of the person on the ground from the base of the tree.

$$\tan 42 = \frac{72}{x}$$

$$x = \frac{72}{\tan 42}$$
$$= 79.96$$

(2dp) ✓ ✓